SPECIFICATION — NODE 5 AS SYSTEM CONTEXT ANCHOR — PAGE 3

SYSTEM-CRITICAL CONTEXT: BASELINES, STATE-CONDITIONED INTERPRETATION, AND LONGITUDINAL COMPARABILITY FOR INTIMATE PHYSIOLOGY (NON-LIMITING)

A. Overview (Non-Limiting)

In various embodiments, Node 5 provides upper-body context that transforms intimate-region sensing (Nodes 1–4) from isolated readings into interpretable physiology by enabling (i) systemic state labeling, (ii) context-conditioned baselines, and (iii) like-for-like longitudinal comparability. Node 5 outputs are used as a control layer that determines how intimate data is interpreted, compared, and trusted over time.

B. Context-Conditioned Baselines (Non-Limiting)

In various embodiments, the computing system establishes one or more baselines for intimate-node outputs conditioned on systemic context derived from Node 5. Non-limiting baseline categories include: 1. rest-state baseline (low motion, stable wear integrity, low activation); 2. sleep baseline (sleep-tagged windows, stable thermal coupling); 3. active-state baseline (activity-tagged windows with appropriate motion thresholds); and 4. elevated activation baseline (autonomic activation tag indicates elevated systemic state). In certain embodiments, intimate-node outputs are compared only within the same baseline category (e.g., rest-to-rest), reducing false drift and improving interpretability.

C. State-Conditioned Interpretation of Intimate Outputs (Non-Limiting)

In various embodiments, Node 5-derived context changes the interpretation of intimate-node outputs by providing state labels and conditioning factors, including: 1. Autonomic conditioning: activation state influences vascular-related interpretations; the system tags intimate outputs as activation-conditioned and adjusts confidence accordingly. 2. Thermal conditioning: temperature context supports plausibility checks and drift awareness for signals influenced by perfusion and tissue temperature. 3. Motion conditioning: high-motion windows are excluded, down-weighted, or labeled as low-confidence to prevent erroneous inference. 4. Wear integrity conditioning: intimate outputs are suppressed, down-weighted, or labeled when Node 5 indicates poor wear integrity, reducing misinterpretation from contact artifacts.

D. Longitudinal Comparability and Like-for-Like Trend Analysis (Non-Limiting)

In various embodiments, Node 5 enables longitudinal analysis by ensuring that trends are computed from comparable systemic states. Non-limiting approaches include: 1. selecting time windows from repeated days that match a target systemic state (e.g., rest-state windows) before comparing intimate metrics; 2. rejecting or reducing influence of windows captured under materially different systemic context (e.g., active vs. rest); 3. computing trend summaries separately by context category; and 4. computing confidence indicators that reflect context consistency across the time series. In certain embodiments, trend outputs are reported as context-separated summaries (e.g., rest-state trend vs active-state trend) with confidence indicators.

E. Systemic Truth Layer for Intimate Signal Ambiguity (Non-Limiting)

In various embodiments, the system treats Node 5 as a systemic truth layer to resolve ambiguity when intimate-local sensors alone cannot determine whether a change is physiologic versus artifact.
Non-limiting examples include: 1. distinguishing physiologic changes from motion-related artifacts when intimate sensor signals shift during movement; 2. disambiguating vascular/tissue-related changes caused

by systemic activation vs local device changes; 3. detecting implausible changes correlated with poor wear integrity rather than physiology; and 4. providing context tags that prevent over-interpretation of low-confidence windows.

F. Multi-Node Context Synchronization and Matched Context Windows (Non-Limiting)

In various embodiments, Node 5 supports selection of matched context windows across nodes, including: 1. matched rest-state windows for Nodes 1 and 2 longitudinal comparisons; 2. matched low-motion windows for Nodes 3 and 4 session outputs; and 3. matched thermal-stable windows for improved plausibility and repeatability. In certain embodiments, correlation or comparison outputs are generated only from matched context windows to improve reliability.

G. Privacy-Preserving Reporting of Context (Non-Limiting)

In various embodiments, Node 5 context is reported as abstract indices and tags (e.g., rest/active, elevated activation, thermal stable/unstable, wear integrity high/low) rather than raw ECG/PPG/EDA waveforms unless explicitly enabled. The system may display confidence labels alongside intimate outputs based on Node 5 context gating and context consistency.

H. Variations

Any embodiment described herein may vary sensor selection, thresholds, baseline definitions, and context categories while maintaining the functional intent of Node 5 providing systemic context needed for state-conditioned interpretation and longitudinal comparability of intimate physiology.